

2016.0 RANGE ROVER (LG), 211-04

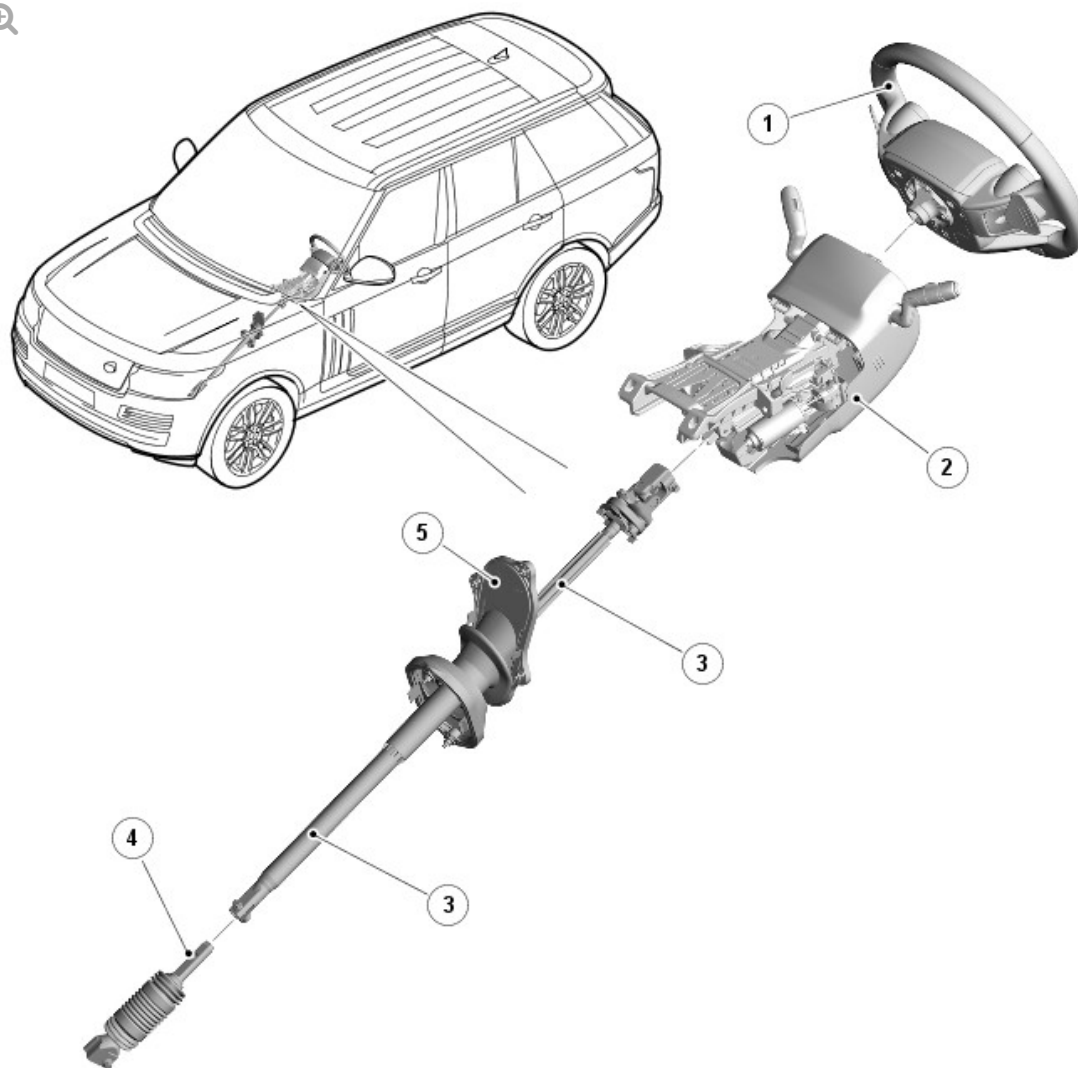
STEERING COLUMN

DESCRIPTION AND OPERATION

COMPONENT LOCATION

NOTE:

LHD (left-hand drive) installation shown, RHD (right-hand drive) installation similar.



E142518

ITEM	DESCRIPTION
1	Steering wheel
2	Upper column
3	Upper shaft
4	Lower shaft
5	Seal assembly

OVERVIEW

The steering column consists of the upper and lower shafts, the upper column and the steering wheel, which are connected together to transfer

driver steering inputs to the steering gear.

The steering column incorporates electric tilt and reach adjustment.

Depending on model and market specifications, the steering column may also include an electric steering column lock and/or a steering wheel heater.

An adjustment switch for tilt and reach adjustment is installed in the lower cowl of the upper column.

A steering wheel module and clockspring assembly, installed on the upper column, incorporates two steering column multifunction switches for control of:

- Exterior lighting.

For additional information, refer to: [Exterior Lighting](#) (417-01 Exterior Lighting, Description and Operation).

- Wipers and washers.

For additional information, refer to: [Wipers and Washers](#) (501-16 Wipers and Washers, Description and Operation).

Two steering wheel switchpacks are installed on the steering wheel for control of:

- Steering wheel heater (where fitted).

- Speed control.

For additional information, refer to: [Speed Control](#) (310-03B Speed Control - V6 S/C 3.0L Petrol, Description and Operation).

- Audio/Video. For additional information, refer to: (415-01 Information and Entertainment System)

[Audio System](#) (Description and Operation),

[Video System](#) (Description and Operation).

- Vehicle information and settings menu.

For additional information, refer to: [Information and Message Center](#) (413-08 Information and Message Center, Description and Operation).

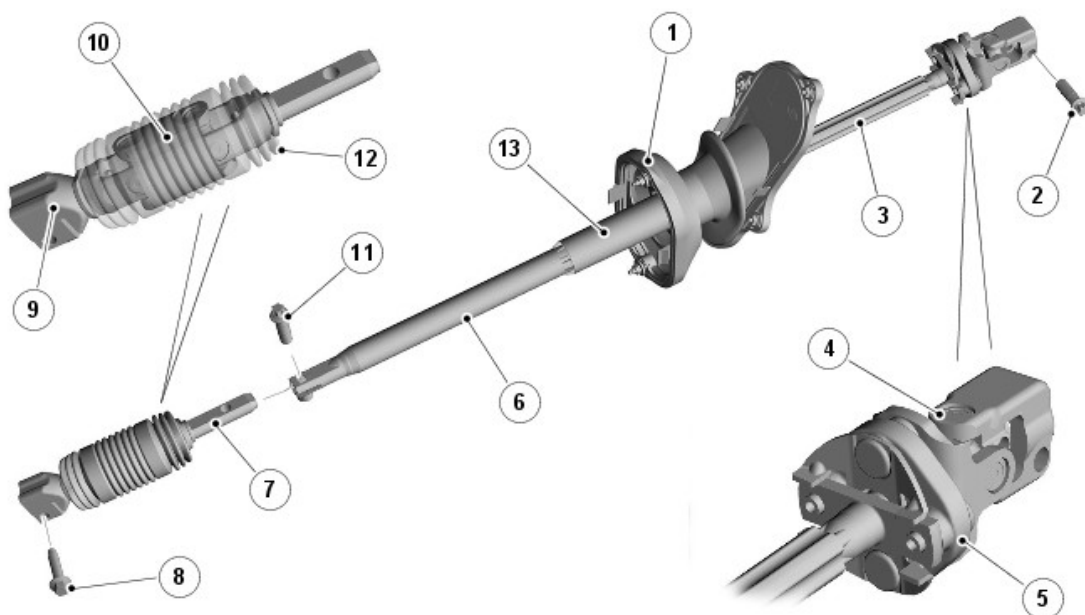
- Telephone and voice control. For additional information, refer to: (415-01 Information and Entertainment System)
[Cellular Phone](#) (Description and Operation),
[Voice Control](#) (Description and Operation).

The steering wheel also incorporates:

- The driver airbag, with integrated horn switch.
For additional information, refer to: [Airbag and Safety Belt Pretensioner Supplemental Restraint System](#) (501-20B Supplemental Restraint System, Description and Operation).
- Two paddle switches for CommandShift™ operation of the transmission.
For additional information, refer to: [External Controls](#) (307-05A Automatic Transmission/Transaxle External Controls - Vehicles With: 8HP45 8-Speed Automatic Transmission AWD, Description and Operation).

DESCRIPTION

UPPER AND LOWER SHAFTS



E142519

ITEM	DESCRIPTION
1	Seal assembly
2	Bolt
3	Inner tube
4	Universal joint
5	Isolator disc
6	Outer tube
7	Stub shaft
8	Bolt
9	Pinion extension
10	Double universal joint
11	Bolt
12	Boot
13	Bearing sleeve

The upper shaft consists of inner and outer tubes that can move telescopically to:

- Permit system assembly.
- Allow steering wheel adjustment.
- Isolate the steering wheel from any movement of the steering gear during an impact.

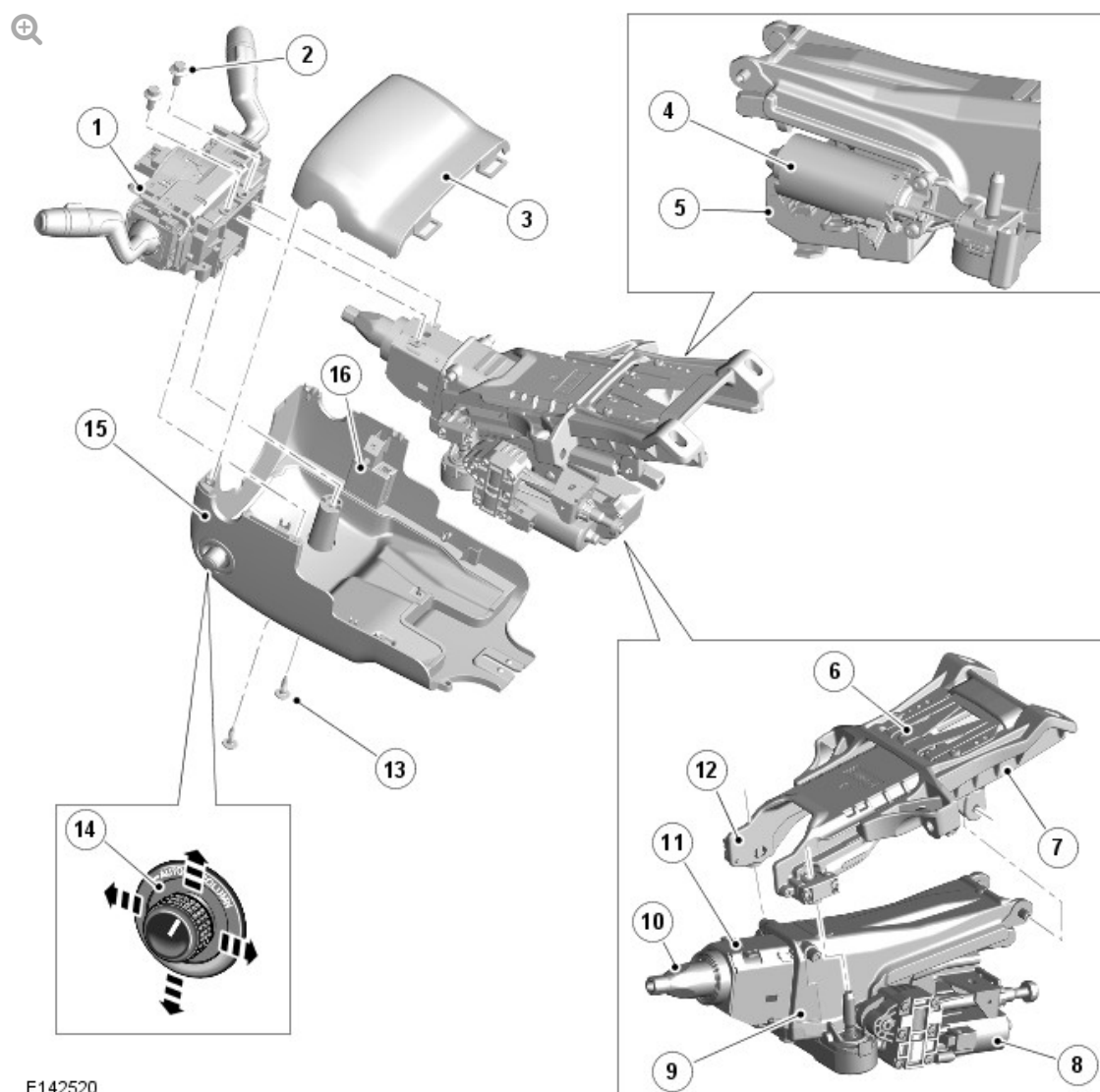
A universal joint connects the inner tube to the upper column. An isolator disc, between the universal joint and the inner tube, reduces NVH (noise, vibration and harshness) in the steering column.

A seal assembly, installed on a bearing sleeve on the outer tube, seals the upper shaft where it passes through the engine and secondary bulkheads. A bearing allows the upper shaft to rotate within the seal assembly.

The lower shaft consists of a double universal joint with a stub shaft and

pinion extension to connect it to the upper shaft and the steering gear respectively. The universal joint is fitted with a corrugated boot.

UPPER COLUMN



E142520

ITEM	DESCRIPTION
1	Steering wheel module and clockspring assembly
2	Screw (2 off)
3	Upper cowl
4	Tilt adjustment motor
5	Electric steering column lock or dummy lock
6	Energy management plate
7	Outer bracket

8	Reach adjustment motor
9	Swing bracket
10	Column spindle
11	Column jacket
12	Inner bracket
13	Screw (2 off)
14	Steering column adjustment switch
15	Lower cowl
16	IAU transceiver (where fitted)

The column spindle is part of the column jacket assembly, that is installed in the swing bracket. An electric motor on the swing bracket drives the column jacket assembly axially within the swing bracket to provide the reach adjustment of the steering column.

The swing bracket is attached to the inner bracket with two pivot pins at the front and two threaded spindles at the rear. A second electric motor on the swing bracket drives the spindles to raise and lower the rear of the swing bracket, relative to the inner bracket, to provide the tilt adjustment of the steering column.

The inner bracket is located on rails in an outer bracket, which is attached to the cross car beam. The inner and outer brackets are locked together by an energy management plate, which is attached to both brackets with screws. In a serious impact, to prevent excessive force causing injury as the driver strikes the airbag, the inner bracket progressively moves forwards on the rails in the outer bracket. The movement is controlled by sections of the energy management plate that peel back in a controlled manner, absorbing energy and allowing the steering wheel to move with the driver.

The steering wheel module and clockspring assembly is attached to the top of the column jacket by two screws. The assembly includes the multifunction switches for the exterior lighting and for the wipers and washers.

Upper and lower cowls cover that part of the upper column not concealed by the instrument panel. The steering column adjustment switch, for control of the tilt and reach adjustment motors, is installed in the right side of the lower cowl. The lower cowl also contains an IAU (immobilizer antenna unit) transceiver for the passive entry passive start system.

For additional information, refer to: [Anti-Theft - Passive](#) (419-01B Anti-Theft - Passive, Description and Operation).

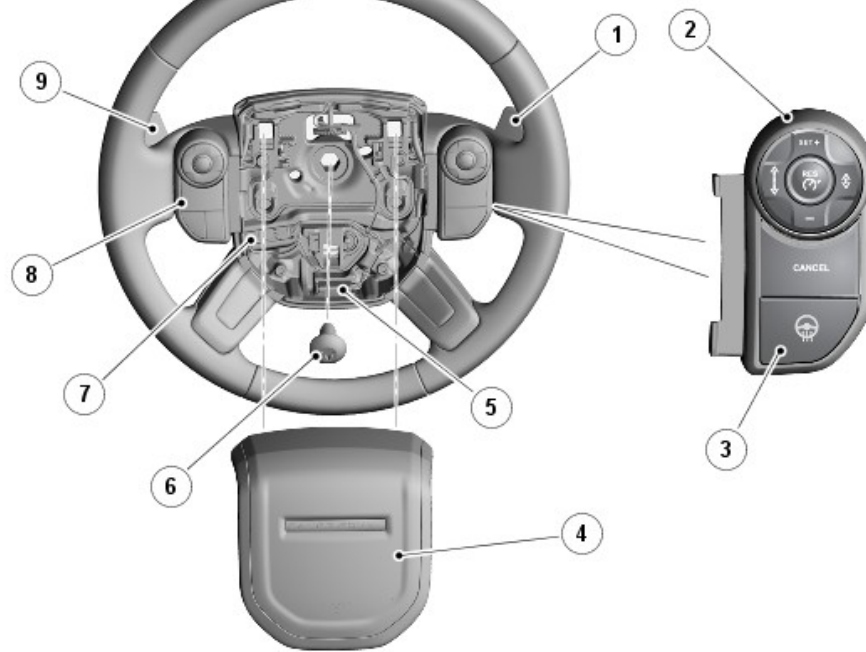
ELECTRIC STEERING COLUMN LOCK (WHERE FITTED)

The electric steering column lock is attached to the underside of the column jacket assembly. The lock engages with a locking sleeve on the shaft of the upper column to lock the steering. A tolerance ring is installed between the locking sleeve and the spindle of the upper column. The tolerance ring allows the spindle to turn in the locking sleeve if a high torque is applied to the steering wheel when the lock is engaged. This prevents the locking bolt from being sheared by someone forcibly turning the steering wheel while the steering lock is engaged, while still effectively locking the steering. Engagement and disengagement of the electric steering column lock is controlled by the passive anti-theft system.

For additional information, refer to: [Anti-Theft - Passive](#) (419-01B Anti-Theft - Passive, Description and Operation).

In markets where an electric steering column lock is not fitted, a dummy lock is installed to provide a forward attachment point for the steering column lower cowl, and the electric steering column lock is disabled in the car configuration files.

STEERING WHEEL



E142521

ITEM	DESCRIPTION
1	Upshift paddle switch
2	Right steering wheel switchpack
3	Steering wheel heater switch (where fitted)
4	Driver airbag
5	Steering wheel heater control module (where fitted)
6	Steering wheel securing screw
7	Heater mat / temperature sensor connector (where fitted)
8	Left steering wheel switchpack
9	Downshift paddle switch

The steering wheel is located on the hexagonal drive end of the column spindle and secured with a M10 pan headed screw. The end of the spindle is marked to indicate the straight ahead position.

The driver airbag and two steering wheel switchpacks are installed in the center of the steering wheel. Upshift and downshift paddle switches are

attached to the front of the steering wheel. A link harness connects the steering wheel components to the clockspring.

STEERING WHEEL HEATER (WHERE FITTED)

The steering wheel heater consists of a heater mat and a NTC (negative temperature coefficient) temperature sensor installed around the rim of the steering wheel under the trim. The heater mat and temperature sensor are connected to a steering wheel heater control module integrated into the link harness under the driver airbag. A switch on the right steering wheel switchpack switches the heater on and off.

OPERATION

STEERING COLUMN ADJUSTMENT

Operation of the steering column adjustment is controlled by the steering column adjustment switch and the CJB (central junction box).

When the steering column adjustment switch is in the COLUMN position, forward, rearward, up and down selections on the switch complete a reference voltage circuit with the CJB, through different elements of a resistive ladder in the switch. The CJB relates the different reference voltages produced to the individual switch selections, then outputs a voltage to the appropriate steering column adjustment motor to adjust the steering column in the required direction. Each steering column adjustment motor incorporates a Hall effect sensor, which allows the CJB to determine the position of the steering column.

When the steering column adjustment switch is in the AUTO position, the switch completes a reference voltage circuit with the CJB that bypasses the resistive ladder. When the CJB detects that the steering column adjustment switch is in the AUTO position, it enables the easy exit/entry mode to provide easier exit from, and entry to, the vehicle. When the driver door is opened, the steering column moves to the uppermost tilt position. When the driver door is closed and the ignition is selected on, the steering column

moves to the driving position associated with the smart key being used.

The memory function for the driver seat also stores the position of the steering column for each memory setting.

For additional information, refer to: Seats (501-10A, Description and Operation).

STEERING WHEEL HEATER (WHERE FITTED)

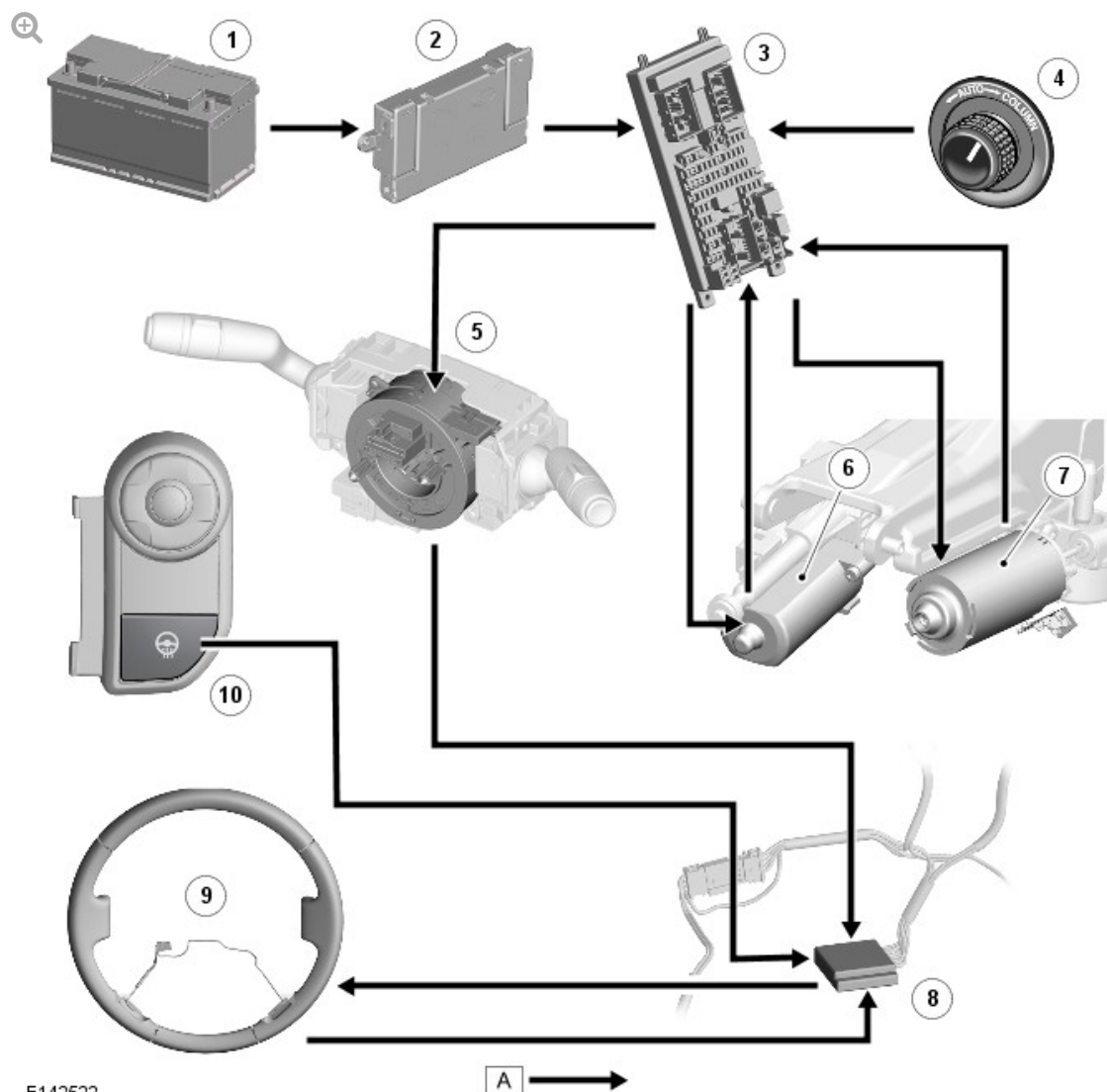
Operation of the steering wheel heater is controlled by the steering wheel heater switch and the steering wheel heater control module.

Power for the steering wheel heater is supplied to the steering wheel heater control module from the ignition relay in the CJB via the clockspring.

Operating the steering wheel heater switch provides a ground signal to the steering wheel heater control module, which then supplies power to the heater mat and the temperature sensor of the steering wheel heater. The steering wheel heater control module monitors the temperature of the heater mat using the input from the temperature sensor, and regulates the power supply to the heater mat accordingly to maintain it at the optimum temperature.

A heater symbol in the switch is illuminated amber when the steering wheel heater is on. When the steering wheel heater is off, the symbol is either not illuminated or, if the exterior lighting is on, illuminated green.

CONTROL DIAGRAM



E142522

ITEM	DESCRIPTION
1	Battery
2	Battery junction box
3	Central junction box
4	Steering column adjustment switch
5	Clockspring
6	Steering column reach adjustment motor
7	Steering column tilt adjustment motor
8	Steering wheel heater control module (where fitted)
9	Steering wheel heater (where fitted)
10	Steering wheel heater switch (where fitted)

